

TOPIC 6: VARIATIONS

1. Direct Variation

y varies directly as x means $y = kx$. Find k from a given pair (x, y), then use $y = kx$ for other values.

2. Inverse (Indirect) Variation

y varies inversely as x means $y = k/x$, or $xy = k$. As x increases, y decreases so that the product stays constant.

3. Joint Variation

y varies jointly as x and z means $y = kxz$. Combine direct variation with two or more variables.

4. Partial Variation

y is partly constant and partly varies as x: $y = k■ + k■x$. Use two given pairs of values to form equations and solve for $k■$ and $k■$.

5. Method

Write the correct formula → substitute known values to find k (or $k■$, $k■$) → substitute again to find the unknown. Always state the relation clearly before calculating.

6. Summary

Direct, inverse, joint and partial variation describe different relationships. The constant of variation links the variables and is found from given data.

End of Topic 6